Buxin Su

Curriculum Vitae

Philadelphia, Pennsylvania — subuxin@sas.upenn.edu — Personal Website — Google Scholar

EDUCATION

Ph.D. in Applied Mathematics and Computational Science 2023.09 - 2027.06
University of Pennsylvania
Advisors: *Weijie Su* and *Bingxin Zhao*

M.S. in Pure Mathematics 2021.09 - 2023.06
University of Pennsylvania
Advisors: *Bingxin Zhao* and *Robin Pemantle*
Thesis: *The exact risks of reference panel-based regularized estimators*

B.S. in Pure Mathematics & **Applied Mathematics Specialist** 2017.09 - 2021.06
University of Toronto
Graduated with High Distinction

AWARDS AND HONORS

NESS Student Research Award 2026.05
New England Statistics Symposium

Two Sigma PhD Fellowship Finalist 2026.01
Two Sigma

Lawrence D. Brown Best Paper Award 2025.10
The Wharton School, University of Pennsylvania

Discussion Paper, Journal of the American Statistical Association 2025.08
American Statistical Association

ICSA NSF Travel Award 2025.07
ICSA 2025 Applied Statistics Symposium

Student Best Paper Award, Statistical Learning and Data Science Section 2024.01
American Statistical Association

RESEARCH INTERESTS

My research lies at the intersection of statistics, machine learning, and the social sciences. I develop statistical theory and methods to address three connected challenges: improving how the AI community evaluates research, increasing the utility of social data under privacy constraints, and making statistical learning more reliable for biomedical applications. My work combines theoretical analysis, empirical experiments, and practical implementation. Contributions include **methods incorporated into the ICML 2026 review process**, methods that **reduce noise variance in the 2020 U.S. Census by about 20%** while maintaining the same level of privacy protection, and studies of the statistical foundations of biomedical prediction. These efforts form three complementary research directions:

Statistical Methods of AI Research Evaluation

- **Using Author Knowledge to Improve Peer Review at AI Conferences.** Developing, evaluating, and implementing statistical methods that incorporate authors' knowledge of their own work, specifically their rankings of their own papers, into peer review at AI/ML conferences [J1, J3, J4, J5]. Recognized through a *JASA* Discussion Paper and coverage in *Nature Index*, this work helps improve research evaluation as AI accelerates the pace of scientific discovery.
- **Randomized Experiment on AI-Assisted Scientific Review.** Conducting a randomized experiment and survey research at ICML 2026, a conference involving over 24,000 papers and 17,000 reviewers, to evaluate how policies on AI assistance affect review behavior and outcomes [C3].

Improving the Utility of Social Data under Privacy Constraints

- **Exact Privacy Accounting for Social Data.** Characterizing the privacy–utility trade-off of the 2020 U.S. Census disclosure-avoidance system through theoretical analysis and showing that it added more noise than necessary [J2]. This analysis shows that noise variances could be reduced by 15.08%–24.82% while maintaining comparable privacy protection. Developing practical methods to determine precisely the noise required for large-scale census releases [P3]. Featured in *New Scientist* under the title "Simple fix could make US census more accurate but just as private."
- **Private Learning with Improved Utility.** Developing theoretical tools and algorithms to improve learning utility while preserving privacy [C1, P5] and to improve predictive performance in federated learning under privacy constraints [C2].

Reliable Statistical Learning for Biomedical Applications

- **Characterizing Prediction Risk in Statistical Learning.** Characterizing the prediction risks of statistical learning methods commonly used in biomedical applications when trained on summary data in high-dimensional settings [P1, P2]. Complementary work examines how synthetic rationale data used in fine-tuning affect disease prediction [P4]. These contributions have received student research awards from the American Statistical Association and the New England Statistics Symposium.

SELECTED PUBLICATIONS

Journal Articles (Published or Accepted)

J1. **Buxin Su**, Jiayao Zhang, Natalie Collina, Yuling Yan, Didong Li, Kyunghyun Cho, Jianqing Fan, Aaron Roth, Weijie Su, "The ICML 2023 Ranking Experiment: Examining Author Self-Assessment in ML/AI Peer Review," *Journal of the American Statistical Association*, 2025. *Link* **[Selected as a Discussion Paper]**

Discussants: Dinghuai Zhang and Yoshua Bengio; Andreas Haupt and Sanmi Koyejo; Xiao-Li Meng; Weichen Wang and Chengchun Shi; Ying Wang and Mengye Ren; Linjun Zhang and Lexin Li.

J2. **Buxin Su**, Weijie Su, and Chendi Wang, "The 2020 United States Decennial Census Is More Private Than You (Might) Think," *Proceedings of the National Academy of Sciences*, 2025. *Link* **[Featured by *New Scientist*]**

J3. **Buxin Su**, Natalie Collina, Garrett G Wen, Didong Li, Kyunghyun Cho, Jianqing Fan, Bingxin Zhao, Weijie Su, "Self-rankings as a predictor of scientific impact beyond peer review," *Nature Computational Science*, 2026. *Link* **[Featured by *Nature Index*, *Nature Computational Science News & Views*, *Wharton AI & Analytics Initiative*, *Penn Today*, and *The Daily Pennsylvanian*]**

J4. **Buxin Su**, Jiayao Zhang, Natalie Collina, Yuling Yan, Didong Li, Kyunghyun Cho, Jianqing Fan, Aaron Roth, and Weijie Su, "Rejoinder: The ICML 2023 Ranking Experiment: Examining Author Self-Assessment in ML/AI Peer Review," *Journal of the American Statistical Association*, 2026. *Link*

J5. Garrett G Wen, **Buxin Su**, Natalie Collina, Zhun Deng, and Weijie Su, "Recommending Best Paper Awards for ML/AI Conferences via the Isotonic Mechanism," *Journal of Machine Learning Research*, accepted, 2026. *Link*

Conference Proceedings

C1. Chendi Wang*, **Buxin Su*** ($*$ = **equal contribution**), Jiayuan Ye, Reza Shokri, and Weijie Su, "Unified Enhancement of Privacy Bounds for Mixture Mechanisms via f-Differential Privacy," *Neural Information Processing Systems (NeurIPS)*, 2023. *Link*

C2. Xiang Li*, **Buxin Su***, Chendi Wang*, Qi Long, and Weijie Su, "Mitigating the Privacy–Utility Trade-off in Decentralized Federated Learning via f-Differential Privacy," *Neural Information Processing Systems (NeurIPS Spotlight)*, 2025. *Link*

C3. Sunnie S. Y. Kim, Wesley Hanwen Deng, Jennifer Wortman Vaughan, **Buxin Su**, Weijie Su, Alekh Agarwal, Sharon Li, Martin Jaggi, Daniel G. Goldstein, Nihar B. Shah, and Miroslav Dudík, "Use and Effects of LLMs in Peer Review: A Randomized Experiment and Survey at ICML 2026," *AI-Native Academia Workshop at NeurIPS 2026 (Oral)*, 2026. *Link*

Preprints

P1. **Buxin Su**, Qiang Sun, Xiaochen Yang, and Bingxin Zhao, "The Exact Risks of Reference Panel-Based Regularized Estimators," under major revision at *Journal of the American Statistical Association*, 2024. *Link* [**Won the 2024 American Statistical Association Student Best Paper Award, Statistical Learning and Data Science Section**]

P2. **Buxin Su**, Jiaoyang Huang, Jin Jin, and Bingxin Zhao, "On Self-Training of Summary Data with Genetic Applications," under major revision at *Journal of the Royal Statistical Society Series B*, 2025. *Link* [**Received the 2026 New England Statistics Symposium Student Research Award**]

P3. **Buxin Su**, Weijie Su, and Chendi Wang, "A Sieve-Accelerated Quadrature Method for Exact Privacy Accounting in the 2020 U.S. Decennial Census," *arXiv:2606.29835*, 2026. *Link*

P4. **Buxin Su**, Bingxuan Li, Cheng Qian, Jin Jin, and Bingxin Zhao, "Supervised Fine-tuning with Synthetic Rationale Data Hurts Real-World Disease Prediction," *arXiv:2606.10279*, 2026. *Link*

P5. **Buxin Su***, Qiaoshi Yang*, Yiding Su*, and Chendi Wang, "Unifying Privacy Accounting: Information Equivalence and Information Loss," *arXiv:2610.02414*, 2026. *Link*

PROFESSIONAL SERVICE

Associate Chair
Forty-Third International Conference on Machine Learning (ICML 2026) *2026.01*

Session Chair
New England Statistics Symposium 2026 *2026.05*
Session: "Modern Statistical Learning, Predictive Inference, and Uncertainty Quantification"

Joint Statistical Meetings 2025 *2025.08*
Session: "Large Language Models in Biomedical and Statistical Knowledge Discovery"

Reviewer
Journals: *IEEE Access, Behaviour & Information Technology, IEEE Journal on Selected Areas in Information Theory*
Conferences: *ICLR 2026*

TEACHING

Teaching Assistant
STAT 1020 *2026.09 - 2026.12*
The Wharton School, University of Pennsylvania

INVITED TALKS

Statistics for AI and Society: Evaluating AI Research and Improving Social Data 2026.10
Statistics Seminar, New Jersey Institute of Technology

On Self-Training of Summary Data with Genetic Applications 2026.05
New England Statistics Symposium 2026

The Three-Year ICML Ranking Experiment: Using Authors' Self-Rankings to Improve Peer Review 2026.01
Two Sigma PhD Fellowship Finalist Reception 2026

The 2020 U.S. Decennial Census Is More Private Than You Think 2025.10
Lawrence Brown Student Workshop 2025

The 2020 U.S. Decennial Census Is More Private Than You Think 2025.06
ICSA Applied Statistics Symposium 2025

The Exact Risks of Reference Panel-Based Regularized Estimators 2024.08
Joint Statistical Meetings 2024

The ICML 2023 Experiment: Can Authors' Opinions on Their Own Submissions Assist Peer Review? 2023.11
Warren Center Lightning Talks, University of Pennsylvania

The Exact Risks of Reference Panel-Based Regularized Estimators 2023.07
ICSA 2023